FieldMesh Custom Firmware

Auto-Advert · GPS Tracking · Off-Grid Mode · SOS System · Message History

Documentation v17 — Based on MeshCore v1.14.1

Hardware: Elecrow ThinkNode M1 (primary) · Seeed Wio Tracker L1 (secondary) · Seeed T1000-E / headless (V5.10)

Covers all changes from V1.00 through V5.2

This document is the complete reference for all deviations from the original MeshCore code. It supersedes all previous documentation. Upload this document with every new MeshCore base version and use it as the sole source for re-applying all patches.

1. Changed Files — Overview (current state V5.2)

The following files deviate from the original MeshCore code. All other files are unchanged.

File	Change
src/helpers/BaseChatMesh.cpp	populateContactFromAdvert() and update block: contact.type stores Bit 4 (ADV_LATLON_MASK=0x10) for GPS flag (V1.05)
examples/companion_radio/AbstractUITask.h	V3: triggerSOS() virtual default (ui-orig compat). V5: newMsg() +is_favorite. V5.2: +channel_name parameter
examples/companion_radio/MyMesh.h	AUTO_ADVERT_INTERVAL_MS, AdvertPath struct, next_auto_advert, setGPSAdvertEnabled(), isOffGridActive(), toggleOffGrid(), sendSOS(), sendChannelMessage() (V5), isSenderFavorite() (V5), _app_initialized (V5.11)
examples/companion_radio/MyMesh.cpp	Constructor, begin(), CMD_SET_OTHER_PARAMS (V3.05), loop(), advert(), sendSOS(), onChannelMessageRecv(), sendChannelMessage() (V5), isSenderFavorite() / extractSenderName() (V5/V5.12), _app_initialized gate (V5.11), updateOnlineNode() calls (V5.2)
examples/companion_radio/NodePrefs.h	4 new fields: normal_freq, normal_bw, normal_sf, normal_cr (off-grid backup)
examples/companion_radio/DataStore.cpp	saveContacts(): contact.type & 0x0F. New fields load/save (bytes 90–99)
examples/companion_radio/ui-new/UITask.h	V4: outdoor_menu, sos_alert, sos_send, triggerSOS(), buzzer, backlight. V5: msg_history, msg_filter, fav_preview, channel_select (L1), msg_compose (L1), _num_unread, _latest_fav_name/_time, setCurrScreen() public, gotoMsgFilter(), gotoChannelSelect(), decrementUnread(), resetUnread(). V5.2: OnlineNode cache, updateOnlineNode(), countOnlineNodes(), countOnlineNodesExcluding(), clearFavoriteCache()
examples/companion_radio/ui-new/UITask.cpp	V1–V4: Haversine, TRACKING, OutdoorMenu, SOSScreens, handleLongPress, backlight. V5: MsgHistoryScreen, MsgFilterScreen, FavPreviewScreen, ChannelSelectScreen (L1), MsgComposeScreen (L1), HomeScreen page 0 rebuild, formatRelTime(), newMsg() rework, msgRead(). V5.2: FIRST page redesign (online counter, mode indicators, fav line), adaptive row limits RECENT/TRACKING, updateOnlineNode() impl., clearFavoriteCache() on sync
examples/companion_radio/ui-new/icons.h	advert_icon_large[] 48×48px (V4.02); nodes_icon[] 8×8px filled circle (V5.2)
examples/companion_radio/ui-orig/Button.h	QUINTUPLE_PRESS in EventType enum; onQuintuplePress() setter; _onQuintuplePress member (V5.10)
examples/companion_radio/ui-orig/Button.cpp	_clickCount == 4 for quadruple; >= 5 for QUINTUPLE_PRESS; triggerEvent() case (V5.10)
examples/companion_radio/ui-orig/UITask.h	handleButtonQuintuplePress(); triggerSOS() override; _sos_active, _buzzer_restore_quiet; newMsg() signature +channel_name (V5.2); refreshDisplay() and isOnRecentOrTrackingPage() no-op overrides (V5.10)
examples/companion_radio/ui-orig/UITask.cpp	Off-Grid toggle 5x; SOS buzzer alarm; forced button feedback; ascending=on, descending=off; _buzzer_restore_quiet pattern (V5.10)
variants/thinknode_m1/variant.h	PIN_BUTTON2=39 corrected (V3.07a)
variants/thinknode_m1/platformio.ini	AUTO_OFF_MILLIS=0 in both companion environments (V1.00)

2. Patch Overview — All Versions V1.00 to V5.2

Version	Title	Changed Files
V1.00	Base: Auto-Advert timer, Tracking page, E-Ink fix	MyMesh.h, MyMesh.cpp, UITask.cpp, platformio.ini
V1.01	GPS display + advert logic (partially reverted)	MyMesh.cpp, UITask.cpp
V1.02	Rollback MyMesh + keep display fix	MyMesh.cpp, UITask.cpp
V1.03	advert() isEnabled() fix	MyMesh.cpp
V1.04	Tracking recv_timestamp — FAILED	MyMesh.cpp
V1.05	has_gps flag — clean solution	BaseChatMesh.cpp, MyMesh.h, MyMesh.cpp, UITask.cpp
V1.06	RECENT/TRACKING split — attempt (replaced by V1.07)	MyMesh.cpp
V1.07	RECENT always, TRACKING only with GPS	MyMesh.cpp
V1.08	Two separate timestamps (basis for V2)	MyMesh.h, MyMesh.cpp, UITask.cpp
V2.01	Off-Grid frequency correction (repeat_freq_ranges)	MyMesh.cpp
V2.02	Off-Grid menu on Tracking page (OutdoorMenuScreen)	NodePrefs.h, DataStore.cpp, MyMesh.h, MyMesh.cpp, UITask.h, UITask.cpp
V2.03	Recommended off-grid settings EU — documentation only	—
V2.04	Flood-Advert in off-grid mode	MyMesh.cpp
V2.05	Compiler fix: sendFlood type ambiguity	MyMesh.cpp
V2.06	contact.type persistence fix	DataStore.cpp
V2.07	contact.type serial fix — root cause resolved	MyMesh.cpp
V3.01	SOS system: send + receive + alarm	AbstractUITask.h, MyMesh.h, MyMesh.cpp, UITask.h, UITask.cpp
V3.05	Timer reset fix + SOS node name display	MyMesh.h, MyMesh.cpp, UITask.cpp
V3.07a	Portability: PIN_BUTTON2 in variant.h, guards	variant.h, UITask.cpp, UITask.h
V3.07b	Alert texts standardised (English)	UITask.cpp
V3.07c	Manual advert send on RECENT page (long press)	UITask.cpp
V3.08	RX restart after toggleOffGrid() — advert() bugfix	MyMesh.cpp
V4.00	UI base layout: splash, relative Y positions, display.height() guards	UITask.cpp
V4.01	Version label v4, all hint texts raised to height()-13	UITask.cpp
V4.02	SOS screens: advert_icon_large 48×48, font fallback, cleanup	UITask.cpp, icons.h
V4.03	AUTO_OFF_MILLIS=0 backlight init fix (M1 boot flash)	UITask.cpp
V4.04	Wio Tracker L1 support: #if UI_HAS_JOYSTICK guards throughout	UITask.cpp, UITask.h, platformio_L1.ini
V5.00	MsgHistoryScreen replaces MsgPreviewScreen; MsgFilterScreen	UITask.cpp, UITask.h, AbstractUITask.h
V5.01	ChannelSelectScreen + MsgComposeScreen (L1 only)	UITask.cpp, UITask.h, MyMesh.h, MyMesh.cpp
V5.02	HomeScreen page 0 rebuild: MSG counter, favorites line	UITask.cpp
V5.03	KEY_BACK for L1 back button; FavPreviewScreen; isSenderFavorite()	UITask.cpp, UITask.h, MyMesh.h, MyMesh.cpp, AbstractUITask.h

Version	Title	Changed Files
V5.04	MsgHistoryScreen: title layout fix, L1 short-press clears (bug)	UITask.cpp
V5.05	Fix L1 short-press clear bug; MsgFilter back entry (M1)	UITask.cpp
V5.06	Timezone offset (TZ_OFFSET_HOURS) — reverted in V5.09	UITask.cpp, platformio.ini
V5.07	FavPreview title layout aligned with MsgHistoryScreen	UITask.cpp
V5.08	FavPreview channel_tag: [CH] consistent; M1 layout polish	UITask.cpp
V5.09	Timezone approach removed; formatRelTime() everywhere; M1 footer trim	UITask.cpp
V5.10	Headless device support: ui-orig Off-Grid 5x click, SOS buzzer alarm, forced button feedback	ui-orig/Button.h, ui-orig/Button.cpp, ui-orig/UITask.h, ui-orig/UITask.cpp
V5.11	BLE app-init race condition fix: _app_initialized gates all async pushes until handshake complete	MyMesh.h, MyMesh.cpp
V5.12	Favourite detection regression fix (channel messages); extractSenderName pipe-delimiter bug fixed	MyMesh.cpp
V5.13	App-sync clears favourite cache: _latest_fav_name/_time reset in msgRead() on msgcount==0	ui-new/UITask.cpp
V5.2	HomeScreen FIRST page redesign: online counter (● X nodes), mode indicators, fav line, MSG:X hidden when connected; RECENT/TRACKING adaptive row limits; channel_name in newMsg(); online node cache	ui-new/UITask.cpp, ui-new/UITask.h, ui-new/icons.h, MyMesh.cpp, AbstractUITask.h, ui-orig/UITask.h

Version	Title	Changed Files
V5.05	Fix L1 short-press clear bug; MsgFilter back entry (M1)	UITask.cpp
V5.06	Timezone offset (TZ_OFFSET_HOURS) — reverted in V5.09	UITask.cpp, platformio.ini
V5.07	FavPreview title layout aligned with MsgHistoryScreen	UITask.cpp
V5.08	FavPreview channel_tag: [CH] consistent; M1 layout polish	UITask.cpp
V5.09	Timezone approach removed; formatRelTime() everywhere; M1 footer trim	UITask.cpp
V5.10	Headless device support: ui-orig Off-Grid 5x click, SOS buzzer alarm, forced button feedback	ui-orig/Button.h, ui-orig/Button.cpp, ui-orig/UITask.h, ui-orig/UITask.cpp

3. V1.xx Patch Details

V1.00 — Base Implementation

■ *Changed files: MyMesh.h, MyMesh.cpp, UITask.cpp, platformio.ini*

Initial implementation of all core features: auto-advert timer (5 min), TRACKING page, BLUETOOTH/SHUTDOWN pages removed, E-Ink fix via platformio.ini.

platformio.ini — add to both companion environments (BLE + USB):

```
-D AUTO_OFF_MILLIS=0
```

Without this flag the E-Ink display turns off after 15 s. The GxEPD driver has an internal CRC check and only updates the panel when content changes.

MyMesh.h — three insertions:

```
// A) After #include BaseChatMesh.h:
#define AUTO_ADVERT_INTERVAL_MS (5UL * 60UL * 1000UL)
// B) AdvertPath struct (base version; has_gps / gps_timestamp added in V1.05 / V1.08):
struct AdvertPath {
uint8_t pubkey_prefix[7];
uint8_t path_len;
char name[32];
uint32_t recv_timestamp;
uint8_t path[MAX_PATH_SIZE];
};
// C) private section after advert_paths[]:
unsigned long next_auto_advert; // 0 = inactive
```

MyMesh.cpp — four locations:

```
// A) Constructor, after dirty_contacts_expiry=0:
next_auto_advert = 0;
// B) begin(), after gps_interval constrain:
// Start timer if advert_loc_policy == ADVERT_LOC_SHARE
// C) CMD_SET_OTHER_PARAMS: final version see V3.05
// D) loop(), before dirty_contacts_expiry block:
if (next_auto_advert && millisHasNowPassed(next_auto_advert)) {
advert();
next_auto_advert = futureMillis(AUTO_ADVERT_INTERVAL_MS);
}
```

UITask.cpp — HomePage enum:

```
enum HomePage { FIRST, RECENT, RADIO, ADVERT, GPS, Count };
// Remove BLUETOOTH and SHUTDOWN. TRACKING is added in V1.05.
```

V1.01 — GPS Display Fix (isEnabled check) + V1.02 — Rollback advert()

■ *Changed files: MyMesh.cpp, UITask.cpp*

V1.01 attempted to call advert() with stored coordinates — reverted in V1.02. The UITask.cpp fix from V1.01 is kept:

```
bool gps_actually_valid = nmea->isEnabled() && nmea->isValid();
strcpy(buf, gps_actually_valid ? "fix" : "no fix");
// sat / pos / alt show '---' when !gps_actually_valid
```

Without the isEnabled() check the NMEA parser stays in 'fix' state even when GPS is disabled in hardware.

V1.03 — advert() isEnabled() Fix

■ *Changed file: MyMesh.cpp*

advert() was sending GPS coordinates even when GPS was disabled in hardware, because isValid() without isEnabled() still returned the last known fix status.

```
bool MyMesh::advert() {
bool send_gps = false;
if (_prefs.advert_loc_policy != ADVERT_LOC_NONE) {
LocationProvider* loc = sensors.getLocationProvider();
if (loc != NULL && loc->isEnabled() && loc->isValid()) send_gps = true;
}
mesh::Packet* pkt = send_gps
```

```
? createSelfAdvert(_prefs.node_name, sensors.node_lat, sensors.node_lon)
: createSelfAdvert(_prefs.node_name);
if (pkt) { sendZeroHop(pkt); return true; }
return false;
}
```

V1.04 — FAILED (documentation only)

Attempted to update recv_timestamp only for GPS adverts. Did not work because contact.gps_lat/lon holds old value for known contacts. Fully replaced by V1.05.

V1.05 — has_gps Flag: Clean Solution

■ *Changed files: BaseChatMesh.cpp, MyMesh.h, MyMesh.cpp, UITask.cpp*

Core idea: Bit 4 (ADV_LATLON_MASK=0x10) from the advert payload is stored in contact.type.

```
// in populateContactFromAdvert() AND update block:
ci.type = parser.getType() | (parser.hasLatLon() ? 0x10 : 0x00);
```

MyMesh.h — AdvertPath struct, add has_gps:

```
struct AdvertPath {
uint8_t pubkey_prefix[7];
uint8_t path_len;
char name[32];
uint32_t recv_timestamp;
uint8_t path[MAX_PATH_SIZE];
bool has_gps; // true if at least one GPS advert received
};
```

V1.06 — RECENT/TRACKING Split: Attempt (replaced by V1.07)

Attempted to skip advert_paths update when no GPS. Problem: node disappeared from RECENT page. Fully replaced by V1.07.

V1.07 — RECENT always, TRACKING only with GPS

recv_timestamp is now always updated (visible on RECENT). has_gps is only set on GPS advert (filter on TRACKING page). Problem: both pages used the same recv_timestamp — resolved in V1.08.

V1.08 — Two Separate Timestamps (basis for V2)

■ *Changed files: MyMesh.h, MyMesh.cpp, UITask.cpp*

MyMesh.h — AdvertPath struct, add gps_timestamp (final version):

```
struct AdvertPath {
uint8_t pubkey_prefix[7];
uint8_t path_len;
char name[32];
uint32_t recv_timestamp; // always updated — for RECENT page
uint8_t path[MAX_PATH_SIZE];
bool has_gps; // true if at least one GPS advert received
uint32_t gps_timestamp; // time of last GPS advert — for TRACKING page
};
```

MyMesh.cpp — onDiscoveredContact() update block:

```
bool gps_in_this_advert = (contact.type & 0x10) != 0;
p->recv_timestamp = getRTCClock()->getCurrentTime(); // ALWAYS
if (gps_in_this_advert) {
p->has_gps = true;
p->gps_timestamp = getRTCClock()->getCurrentTime(); // GPS only
}
```

UITask.cpp — add TRACKING to HomePage enum, remove ADVERT. Insert Haversine formula (calcDistance function) directly after #include icons.h.

4. V2.xx Patch Details

V2.01 — Off-Grid Frequency Correction (repeat_freq_ranges)

■ *Changed file: examples/companion_radio/MyMesh.cpp*

Band	Before (old)	After (new)
EU 433 MHz	433.000 MHz (illegal, single frequency)	433.050 – 434.790 MHz
EU 869 MHz	869.000 MHz (0.1% duty cycle — unusable)	869.400 – 869.650 MHz (Sub-band g1, 10% DC)
US/AU 915 MHz	918.000 MHz (outside band)	915.000 – 928.000 MHz

```
static FreqRange repeat_freq_ranges[] = {
{ 433050, 434790 }, // EU 433 MHz ISM band
{ 869400, 869650 }, // EU 869 MHz sub-band g1 (no LBT required)
{ 915000, 928000 } // US/AU 915 MHz
};
```

V2.02 — Off-Grid Menu on Tracking Page (OutdoorMenuScreen)

■ *Changed files: NodePrefs.h, DataStore.cpp, MyMesh.h, MyMesh.cpp, UITask.h, UITask.cpp*

Replaces the simple GPS-share toggle with a multi-step outdoor settings menu. Long press on the Tracking page opens the menu.

Menu Entry	Function	State
GPS Share	Auto-advert every 5 min with GPS	[ON] / [OFF]
Off-Grid	Switch to 869.4625 MHz / BW125 / SF11 / CR5 + client_repeat	[ON] / [OFF]
Back	Close menu, return to Tracking page	—

NodePrefs.h — append to end of struct:

```
float normal_freq; float normal_bw; uint8_t normal_sf; uint8_t normal_cr;
```

DataStore.cpp — bytes 90–99, after rx_boosted_gain (byte 89):

```
// Save:
success = success && (file.write((uint8_t*)&_prefs.normal_freq, 4) == 4);
success = success && (file.write((uint8_t*)&_prefs.normal_bw, 4) == 4);
success = success && (file.write(&_prefs.normal_sf, 1) == 1);
success = success && (file.write(&_prefs.normal_cr, 1) == 1);
// Load (backwards-compatible):
file.read((uint8_t*)&_prefs.normal_freq, 4);
file.read((uint8_t*)&_prefs.normal_bw, 4);
file.read(&_prefs.normal_sf, 1);
file.read(&_prefs.normal_cr, 1);
```

MyMesh.h — isOffGridActive() and toggleOffGrid() as new public methods.

MyMesh.cpp — toggleOffGrid() implementation:

```
#define OFFGRID_FREQ 869.4625f
#define OFFGRID_BW 125.0f
#define OFFGRID_SF 11
#define OFFGRID_CR 5

void MyMesh::toggleOffGrid() {
if (_prefs.client_repeat == 0) {
_prefs.normal_freq = _prefs.freq; _prefs.normal_bw = _prefs.bw;
_prefs.normal_sf = _prefs.sf; _prefs.normal_cr = _prefs.cr;
_prefs.freq = OFFGRID_FREQ; _prefs.bw = OFFGRID_BW;
_prefs.sf = OFFGRID_SF; _prefs.cr = OFFGRID_CR;
_prefs.client_repeat = 1;
} else {
if (_prefs.normal_freq > 0.0f) {
_prefs.freq = _prefs.normal_freq; _prefs.bw = _prefs.normal_bw;
_prefs.sf = _prefs.normal_sf; _prefs.cr = _prefs.normal_cr;
}
_prefs.client_repeat = 0;
```

```
}
radio_set_params(_prefs.freq, _prefs.bw, _prefs.sf, _prefs.cr);
savePrefs();
advert(); // V3.08: RX restart via normal TX->RX cycle
}
```

■ advert() at the end is the V3.08 fix — do not remove! See V3.08 for details.

V2.03 — Recommended Off-Grid Settings EU (documentation only, no code)

Parameter	Value
Frequency	869.4625 MHz (channel 1/2 in sub-band g1 at BW 125)
Bandwidth (BW)	125 kHz
Spreading Factor	11
Coding Rate (CR)	5 (= 4/5)
TX power	max. 25 mW ERP
Duty cycle	max. 10%
LBT required?	No

■ The 865–868 MHz band requires LBT (Listen Before Talk) and AAG. MeshCore does not implement these — this band CANNOT be used.

V2.04 — Flood-Advert in Off-Grid Mode / V2.05 — Compiler Fix

In off-grid mode (client_repeat=1) advert() sends a flood advert instead of ZeroHop:

```
if (pkt) {
if (_prefs.client_repeat) {
sendFlood(pkt, (uint32_t)0, _prefs.path_hash_mode + 1); // V2.05: uint32_t cast
} else {
sendZeroHop(pkt);
}
}
```

V2.04 had a compiler error 'sendFlood ambiguous'. V2.05 fixes it with explicit cast: (uint32_t)0. allowPacketForward() also returns true when client_repeat != 0.

V2.06 — contact.type Persistence Fix

■ *Changed file: examples/companion_radio/DataStore.cpp*

```
// BEFORE:
success = success && (file.write(&c.type;, 1) == 1);
// AFTER (V2.06):
uint8_t clean_type = c.type & 0x0F; // do not persist bit 4
success = success && (file.write(&clean;_type, 1) == 1);
```

V2.07 — contact.type Serial Fix (root cause resolved)

■ *Changed file: examples/companion_radio/MyMesh.cpp*

Problem: writeContactRespFrame() was sending contact.type with bit 4 unfiltered to the companion app. Result: app received type=0x11 instead of type=0x01 and did not recognise the contact type — messaging and telemetry not possible.

```
// Fix 1 — writeContactRespFrame():
out_frame[i++] = contact.type & 0x0F; // do not send bit 4 to app
// Fix 2 — updateContactFromFrame() (guard for reverse direction):
contact.type = frame[i++] & 0x0F; // do not accept bit 4 from app
```

5. V3.xx Patch Details

V3.01 — SOS System (Send + Receive + Alarm)

■ *Changed files: AbstractUITask.h, MyMesh.h, MyMesh.cpp, UITask.h, UITask.cpp*

Implements a complete SOS emergency system. Page order after V3: FIRST > RECENT > TRACKING > RADIO > GPS > SOS. The ADVERT page is removed.

AbstractUITask.h — compatibility stub:

```
virtual void triggerSOS(const char* from, const char* text) {} // V3: default empty
```

MyMesh.cpp — sendSOS() implementation:

Searches case-insensitively for a channel whose name contains 'sos' and sends the message there.

```
bool MyMesh::sendSOS() {
for (int i = 0; i < MAX_GROUP_CHANNELS; i++) {
ChannelDetails ch;
if (!getChannel(i, ch)) continue;
char name_lower[33];
for (int j = 0; j < 32 && ch.name[j]; j++)
name_lower[j] = tolower(ch.name[j]);
name_lower[32] = 0;
if (strstr(name_lower, "sos") == NULL) continue;
char text[80];
LocationProvider* loc = sensors.getLocationProvider();
if (loc && loc->isEnabled() && loc->isValid()) {
snprintf(text, sizeof(text), "!SOS %.4f,%.4f",
loc->getLatitude()/1000000.0, loc->getLongitude()/1000000.0);
} else {
strncpy(text, "!SOS no GPS", sizeof(text));
}
uint32_t ts = getRTCClock()->getCurrentTimeUnique();
return sendGroupMessage(ts, ch.channel, _prefs.node_name, text, strlen(text));
}
return false;
}
```

MyMesh.cpp — !SOS detection in onChannelMessageRecv():

Insert this block BEFORE the normal newMsg() call:

```
// V3: SOS detection — strstr because MeshCore prepends node name
if (strstr(text, "!SOS") != NULL) {
if (_ui) _ui->triggerSOS(channel_name, text);
return; // do NOT call newMsg()
}
```

■ *strstr() instead of strncmp() because MeshCore prepends the node name on receipt. Format: 'NodeName: !SOS lat,lon'*

Buzzer tones (notify function):

Event	RTTTL melody	When
contactMessage	MsgRcv3:d=4,o=6,b=200:32e,32g,32b,16c7	Direct message received
channelMessage	kerplop:d=16,o=6,b=120:32g#,32c#	Channel message received
ack	ack:d=32,o=8,b=120:c	Menu action confirmed
SOS alarm	siren:d=8,o=5,b=100:d,e,d,e,d,e,d,e	SOS received (loops continuously)

■ *The SOS siren does NOT run via notify() but directly via triggerSOS() and playSOSAlarm() — so it can override quiet mode.*

SOS system behaviour overview:

Situation	Behaviour
Channel 'sos' not present	Alert 'No SOS channel!' — no crash
GPS not available when sending	Sends '!SOS no GPS'
GPS available when sending	Sends '!SOS lat,lon' (4 decimal places)

Situation	Behaviour
!SOS received, phone connected	Turn on display anyway, show alarm
!SOS received, buzzer on quiet	Turn buzzer on — remains ON after acknowledgement
Normal message during alarm	Ignored — SOS screen has priority
Buzzer tone ends	poll() detects isPlaying()==false, restarts

V3.05 — Timer Reset Fix + SOS Node Name Display

■ *Changed files: MyMesh.h, MyMesh.cpp, UITask.cpp*

Two independent fixes: (1) The auto-advert timer was reset to +5 minutes on every phone connect or display toggle, even when it was already running. (2) The SOS alarm screen was showing the channel name ('#sos') instead of the node name of the sending node.

Fix 1 — Timer reset bug: MyMesh.h (setGPSAdvertEnabled) and MyMesh.cpp (CMD_SET_OTHER_PARAMS):

```
// AFTER (V3.05) — applies to both locations:
if (enabled) {
if (next_auto_advert == 0) { // only start timer if not already running
advert();
next_auto_advert = futureMillis(AUTO_ADVERT_INTERVAL_MS);
}
} else {
next_auto_advert = 0;
}
```

■ *Behaviour: OFF->ON = immediate advert + timer starts. ON->OFF = timer stops. Timer already running = no reset, no duplicate advert.*

Fix 2 — SOS node name: UITask.cpp (SOSAlertScreen::setAlert()):

```
void setAlert(const char* from, const char* text) {
strncpy(_from, from, sizeof(_from)-1); _from[sizeof(_from)-1] = 0;
strncpy(_text, text, sizeof(_text)-1); _text[sizeof(_text)-1] = 0;
// Extract node name from text — everything before the first ':'
_node_name[0] = 0;
const char* colon = strchr(text, ':');
if (colon && colon > text) {
int len = colon - text; if (len > 31) len = 31;
strncpy(_node_name, text, len); _node_name[len] = 0;
for (int i = len-1; i >= 0 && _node_name[i] == ' '; i--) _node_name[i] = 0;
} else {
strncpy(_node_name, from, sizeof(_node_name)-1);
_node_name[sizeof(_node_name)-1] = 0;
}
}
```

V3.07a — Code Cleanup (Portability)

■ *Changed files: variants/thinknode_m1/variant.h, UITask.cpp, UITask.h*

Goal: make code portable for other devices. No functional change on ThinkNode M1.

```
// variant.h — BEFORE:
#define PIN_BUTTON2 (11)
// variant.h — AFTER:
#define PIN_BUTTON2 (39) // P1.07 (corrected per Elecrow datasheet)
```

UITask.cpp — the hack block at the start of the file is fully removed.

```
// UITask.cpp — in loop():
// BEFORE: #ifdef DISP_BACKLIGHT
// AFTER: #if defined(DISP_BACKLIGHT) && !defined(BACKLIGHT_BTN)
```

All three locations where PIN_BUTTON2 is used are now wrapped in #ifdef PIN_BUTTON2.

Device / flag combination	Backlight toggle	Auto-off	Conflict
ThinkNode M1 (DISP_BACKLIGHT + PIN_BUTTON2)	Upper button (GPIO 39)	5 min	None

Device / flag combination	Backlight toggle	Auto-off	Conflict
Device with BACKLIGHT_BTN (e.g. Techo)	Original system	Original	None (guard)
Device with DISP_BACKLIGHT, no PIN_BUTTON2	No toggle button (auto-off only)	5 min	None (guard)
Device without DISP_BACKLIGHT	Not active	Not active	Not active

V3.07b — Alert Texts Standardised (English)

Before	After
"SOS gesendet!"	"SOS sent!"
"Kein SOS-Channel!"	"No SOS channel!"

V3.07c — Manual Advert Send (RECENT page)

■ *Changed file: UITask.cpp*

Long press on the RECENT page immediately sends an advert. advert() internally decides ZeroHop vs. Flood based on _prefs.client_repeat.

```
// 1. RECENT page render(): add hint line at bottom
display.drawTextCentered(..., "Advert: PRESS_LABEL");
// 2. HomeScreen: add isOnRecentPage() method (analogous to isOnTrackingPage)
// 3. handleLongPress(): new branch for RECENT page:
} else if (curr == home && ((HomeScreen*)home)->isOnRecentPage()) {
the_mesh.advert();
notify(UIEventType::ack);
showAlert("Advert sent!", 1500);
_next_refresh = 0; // immediate refresh so alert appears instantly
c = 0;
```

■ *_next_refresh = 0 is necessary because handleLongPress() sets c to 0 before the downstream _next_refresh=100 block fires.*

V3.08 — RX Restart after toggleOffGrid() (retrospective bugfix)

■ *Changed file: examples/companion_radio/MyMesh.cpp*

After toggleOffGrid() the receiver was deaf until the next transmitted packet. The RX log remained empty.

Root cause: radio_set_params() calls RadioLib directly, bypassing the RadioLibWrapper state. The wrapper internally manages STATE_RX/TX/IDLE and on the T1000-E also the RF switch state. A direct radio.startReceive() call in target.cpp is not sufficient — the wrapper state remains wrong.

Fix: advert() at the end of toggleOffGrid(). The normal TX cycle runs completely through the wrapper stack — after TX the wrapper sets state correctly and starts startReceive() internally. RF switch and AGC are handled correctly in the process.

```
void MyMesh::toggleOffGrid() {
// ... parameter logic unchanged ...
radio_set_params(_prefs.freq, _prefs.bw, _prefs.sf, _prefs.cr);
savePrefs();
advert(); // V3.08: RX restart via normal TX->RX cycle of the wrapper
}
```

6. V4.xx Patch Details

All V4.xx changes affect display rendering only (UITask.cpp and icons.h). No logic, function or class structure changes. All layouts use display.height() and display.width() — never absolute pixels. Device-dependent layouts are distinguished at runtime: if (display.height() >= 128) for large displays (M1), otherwise compact layout.

V4.00 — UI Base Layout: Splash, Relative Y Positions, display.height() Guards

SplashScreen — two-group layout (large displays only):

```
if (display.height() >= 128) {
// Group top
display.drawXbm(..., meshcore_logo, 128, 13); // Y=4
display.setTextSize(2); display.drawTextCentered(..., 22, _version_info);
display.setTextSize(1); display.drawTextCentered(..., 42, FIRMWARE_BUILD_DATE);
// Group bottom
display.setTextSize(3); display.drawTextCentered(..., display.height()-42, "FieldMesh");
display.setTextSize(2); display.drawTextCentered(..., display.height()-20, "v4");
}
```

All hint texts — relative Y position:

All status hints showing actions (e.g. 'Settings: long press', 'Advert: long press', 'short=next long=select') were moved from hardcoded values to display.height() - 13. The value -13 (instead of -11) gives descenders (g, p, y) sufficient space.

Screen	Before	After
RECENT hint	64 - 11	display.height() - 13
TRACKING hint	64 - 11	display.height() - 13
OutdoorMenuScreen footer	57 (hardcoded)	display.height() - 13
SOS page hint	54 (hardcoded)	display.height() - 13
SOSAlertScreen hint	57 (hardcoded)	display.height() - 13
SOSSendScreen hints	54 / 57 (hardcoded)	display.height() - 13 / -24

V4.01 — Version Label 'v4', Hint Texts Raised to -13

Version string in splash screen updated from 'v3' to 'v4'. All hint texts finally set to display.height() - 13 (some were still on -11 in V4.00).

V4.02 — SOS Screens: Larger Icon, Layout Corrections, Font Fallback

icons.h — advert_icon_large 48×48 (new):

The new icon advert_icon_large (48×48px) was added as its own bitmap array in icons.h. The original advert_icon (32×32) remains unchanged. The large icon is used exclusively on displays with height >= 128.

■ Important for future icon creation: GIMP exports XBM with MSB-first bit order, GxEPD reads LSB-first. Each byte must therefore be bit-reversed. Conversion: int(format(byte, '08b')[::-1], 2)

SOSAlertScreen — layout and font fallback:

```
// Node name with size fallback:
display.setTextSize(strlen(_node_name) >= 15 ? 1 : 2);
display.setColor(DisplayDriver::YELLOW);
display.drawTextCentered(display.width()/2, 72, _node_name);
// At size 2, approx 1 char per ~12px.
// At 200px display width: max ~16 chars. Threshold 15 leaves buffer for wide chars.
```

V4.03 — AUTO_OFF_MILLIS=0 Backlight Init Fix (M1 Boot Flash)

■ *Changed file: UITask.cpp*

When AUTO_OFF_MILLIS=0 is set (required for E-Ink), the auto-off block is compiled out entirely. However the backlight GPIO was never initialised to LOW at boot, causing the backlight to flash briefly on power-on before the first button press could turn it off.

Fix — in UITask::loop(), inside the DISP_BACKLIGHT block:

```
#if defined(DISP_BACKLIGHT) && !defined(BACKLIGHT_BTN)
// Initialise backlight GPIO to LOW on first loop pass.
if (!_backlight_initialized) {
digitalWrite(DISP_BACKLIGHT, LOW);
_backlight_initialized = true;
}
#endif
```

■ This fix only applies when AUTO_OFF_MILLIS=0 AND DISP_BACKLIGHT is defined AND BACKLIGHT_BTN is not defined. This is exactly the ThinkNode M1 configuration.

V4.04 — Wio Tracker L1 Support

■ *Changed files: UITask.cpp, UITask.h, platformio_L1.ini*

Adds second supported hardware: Seeed Wio Tracker L1 with OLED display and joystick input. All L1-specific code is wrapped in #if UI_HAS_JOYSTICK ... #endif guards.

Feature	M1 (ThinkNode)	L1 (Wio Tracker)
Display	200×200 E-Ink	128×64 OLED
Input	1 button (long/short/double/triple)	Joystick (up/down/left/right/press/long) + back button
GPS	Yes (ENV_INCLUDE_GPS=1)	No
Buzzer	Yes (PIN_BUZZER=6)	No
PRESS_LABEL macro	"long press"	"press Enter"
Compile guard	default (no guard)	#if UI_HAS_JOYSTICK

7. V5.xx Patch Details

V5 adds two standalone features that work without a smartphone. All V5.00–V5.09 changes are confined to UITask.cpp and UITask.h (plus a small wrapper in MyMesh.h/MyMesh.cpp and one signature change in AbstractUITask.h). DataStore.cpp, NodePrefs.h, icons.h remain unchanged.

Feature	Devices	Description
Message Page (page 0 rebuild)	All	Full message history with unread counter and favorites line replaces MSG-count splash
MsgHistoryScreen	All	Ring buffer (32 entries), one message per view, All / Favorites filter
MsgFilterScreen	All	Menu: filter selection (All / Favorites) + Send entry (L1) + Back
FavPreviewScreen	All	Auto-popup for messages from favorite contacts
ChannelSelectScreen	L1 only	Channel selection before composing a message
MsgComposeScreen	L1 only	4-row QWERTY keyboard for text input and send

V5 Architecture Overview

File	Change
AbstractUITask.h	newMsg() signature extended: bool is_favorite = false added as last parameter
MyMesh.h	New public: sendChannelMessage(), isSenderFavorite(), extractSenderName()
MyMesh.cpp	Implementations of the three new methods; onChannelMessageRecv() passes is_favorite to newMsg()
UITask.h	5 new UIScreen* members (3 behind #if UI_HAS_JOYSTICK); 3 new private members (_num_unread, _latest_fav_name, _latest_fav_time); setCurrScreen() moved from private to public; new public helpers: gotoMsgFilter(), gotoChannelSelect(), decrementUnread(), resetUnread()
UITask.cpp	MsgPreviewScreen deleted; MsgHistoryScreen, MsgFilterScreen, FavPreviewScreen added; ChannelSelectScreen + MsgComposeScreen added (#if UI_HAS_JOYSTICK); HomeScreen page FIRST rebuilt; newMsg() reworked; msgRead() updated; formatRelTime() added

What does NOT change: DataStore.cpp/.h, NodePrefs.h, icons.h, all existing UIScreen classes (HomeScreen other pages, OutdoorMenuScreen, all SOS screens), the full offline_queue / contact / advert machinery in MyMesh.

V5 — MsgEntry Data Structure (RAM)

```
// V5 MsgEntry (in MsgHistoryScreen):
struct MsgEntry {
uint32_t timestamp; // RTC time of receipt
char from_name[33]; // sender name
char channel_tag[8]; // "[CH]", "[DM]"
char msg[80]; // message text (80 char limit)
bool is_favorite; // was sender a favorite at time of receipt?
};
#define MAX_HISTORY_MSGS 32
// RAM: 32 x 126 = 4032 bytes
```

■ *is_favorite is determined once at receipt time (lookupContact → contact.flags & 0x01). Not re-evaluated at runtime.*

V5 — newMsg() Rework

V4.04 called setCurrScreen(msg_preview) on every new message — forcing an automatic screen switch. V5 removes this: messages are stored silently in the ring buffer. The screen does not change. Only FavPreviewScreen is shown automatically (favorites only).

```
void UITask::newMsg(uint8_t path_len, const char* from_name,
const char* text, int msgcount, bool is_favorite) {
_msgcount = msgcount;
```

```
if (curr == sos_alert) return; // V3: do not interrupt SOS alarm
_num_unread++;
((MsgHistoryScreen*)msg_history)->addMessage(path_len, from_name, text, is_favorite);
if (is_favorite) {
_latest_fav_time = getRTCClock()->getCurrentTime();
strncpy(_latest_fav_name, from_name, sizeof(_latest_fav_name)-1);
if (curr == msg_history || curr == msg_filter) {
showAlert("+1 fav", 1500);
} else {
((FavPreviewScreen*)fav_preview)->setMessage(path_len, from_name, text);
setCurrScreen(fav_preview);
}
} else if (curr == msg_history) {
showAlert("+1 new", 1500);
}
}
```

V5 — sendChannelMessage() Wrapper (MyMesh)

```
// MyMesh.h — public block:
bool sendChannelMessage(uint8_t channel_idx, const char* text);
// MyMesh.cpp:
bool MyMesh::sendChannelMessage(uint8_t channel_idx, const char* text) {
if (text == nullptr || text[0] == 0) return false;
ChannelDetails ch;
if (!getChannel(channel_idx, ch)) return false;
uint32_t ts = getRTCClock()->getCurrentTimeUnique();
return sendGroupMessage(ts, ch.channel, _prefs.node_name, text, strlen(text));
}
```

V5 — isSenderFavorite() (MyMesh)

Channel messages do not carry a ContactInfo object. The sender name is embedded in the text as 'NodeName | Channel: text'. isSenderFavorite() extracts the name and searches the contact list.

```
bool MyMesh::isSenderFavorite(const char* text) {
char sender_name[33];
if (!extractSenderName(text, sender_name, sizeof(sender_name))) return false;
ContactInfo contact;
for (uint32_t i = 0; i < MAX_CONTACTS; i++) {
if (!getContactByIdx(i, contact)) break;
if (contact.name[0] == 0) continue;
if (strcmp(contact.name, sender_name) == 0)
return (contact.flags & 0x01) != 0; // bit 0 = favorite
}
return false;
}
```

■ *Name collision (two contacts same name): first match wins — accepted risk.*

V5 — formatRelTime() Helper

```
static void formatRelTime(char* buf, size_t buf_size, uint32_t now, uint32_t then) {
int secs = (int)(now - then);
if (secs < 0) secs = 0;
if (secs < 60) snprintf(buf, buf_size, "%ds", secs);
else if (secs < 3600) snprintf(buf, buf_size, "%dm", secs / 60);
else snprintf(buf, buf_size, "%dh", secs / 3600);
}
```

■ *No timezone logic — purely relative elapsed time. Used in: HomeScreen FIRST page, MsgHistoryScreen, FavPreviewScreen.*

V5 — Dual Unread Counter Design

Counter	Location	Owner	Controlled by
_msgcount	UITask member	Synced with MyMesh offline_queue_len	MyMesh via newMsg() / msgRead()

Counter	Location	Owner	Controlled by
_num_unread	UITask member (new V5)	UI-internal display copy	UI navigation (decrementUnread / resetUnread)

■ *The LED blink handler (userLedHandler) uses _msgcount — not _num_unread. LED follows phone sync state, display counter follows UI navigation. Both can diverge. Accepted design decision.*

8. V5.10 Patch Details — Headless Device Support

■ *Changed files: ui-orig/Button.h, ui-orig/Button.cpp, ui-orig/UITask.h, ui-orig/UITask.cpp*

Adds full FieldMesh feature support for devices without a display and with a single button (Seeed T1000-E, RAK WiseMesh TAG and compatible devices). All changes are confined to the ui-orig layer — ui-new (ThinkNode M1, Wio Tracker L1) is completely unaffected. The ui-orig UITask now implements the full AbstractUITask V5 interface.

Button.h — QUINTUPLE_PRESS

Three additions to the upstream file:

- QUINTUPLE_PRESS added to EventType enum (after QUADRUPLE_PRESS, before LONG_PRESS)
- onQuintuplePress(EventCallback) setter added to public interface
- _onQuintuplePress = nullptr member added to private section

Button.cpp — 5x click detection

The upstream file used >= 4 for quadruple detection, which meant 5+ clicks also triggered QUADRUPLE_PRESS. Fixed to == 4, with a new >= 5 branch for QUINTUPLE_PRESS.

```
// BEFORE (upstream):
} else if (_clickCount >= 4) {
triggerEvent(QUADRUPLE_PRESS);

// AFTER (V5.10):
} else if (_clickCount == 4) { // exactly 4
triggerEvent(QUADRUPLE_PRESS);
} else if (_clickCount >= 5) { // 5x click
triggerEvent(QUINTUPLE_PRESS);
}
```

■ The _clickCount == 4 fix is critical — without it, 5x press silently triggers quadruple instead of quintuple.

UITask.h — new members and overrides

Member / Method	Purpose
bool _sos_active	SOS alarm active — buzzer loops until short press acknowledges
bool _buzzer_restore_quiet	Deferred quiet restore after forced tone playback
handleButtonQuintuplePress()	Off-Grid toggle handler (private)
triggerSOS() override	SOS alarm via buzzer — overrides AbstractUITask default no-op
refreshDisplay() override {}	Explicit no-op — MyMesh calls this on headless; ignored
isOnRecentOrTrackingPage() override	Returns false — no pages on headless device
newMsg() with is_favorite param	Updated signature for V5 AbstractUITask compatibility

UITask.cpp — button mapping

Press	Action	Buzzer feedback
1×	Wake display / acknowledge SOS alarm	None (SOS: siren stops)
2×	Send advert (ZeroHop normal, Flood in Off-Grid)	Double beep (c,p,c) — forced
3×	Buzzer mute on/off	Ascending (c,e) = on; descending (e,c) = off — forced
4×	GPS on/off	Ascending (c,e) = on; descending (e,c) = off — forced
5×	Off-Grid mode on/off	Ascending (c,e) = on; descending (e,c) = off — forced
Long (first 8 s)	CLI Rescue Mode	None
Long	Shutdown	None

_buzzer_restore_quiet pattern

buzzer.play() is non-blocking. If buzzer.quiet() is called immediately after play(), the tone is silenced before it finishes. The _buzzer_restore_quiet flag defers the quiet restore until loop() detects buzzer.isPlaying() == false.

```
// Handler:
buzzer.quiet(false);
buzzer.play("...");
_buzzer_restore_quiet = true;

// loop():
if (buzzer.isPlaying()) {
buzzer.loop();
} else {
if (_buzzer_restore_quiet) {
buzzer.quiet(_node_prefs->buzzer_quiet);
_buzzer_restore_quiet = false;
}
if (_sos_active) {
buzzer.play("siren:d=8,o=5,b=100:d,e,d,e,d,e,d,e");
}
}
```

■ Exception: triggerSOS() does NOT set _buzzer_restore_quiet. SOS overrides quiet mode permanently until acknowledged — _node_prefs->buzzer_quiet is set to 0 but intentionally not saved to flash.

SOS alarm behaviour (headless)

Situation	Behaviour
!SOS received	buzzer.quiet(false), play siren, _sos_active = true
Buzzer finishes siren	loop() restarts siren (while _sos_active == true)
Short press	_sos_active = false, buzzer.shutdown() — alarm stops
Long press during SOS	Shutdown (SOS does not block long press on headless)

9. Checklist for New MeshCore Base Version

Work through each of these check points after a base update:

What to check	Why / Where
BaseChatMesh.cpp: contact.type | 0x10	populateContactFromAdvert() + update block — GPS flag lost otherwise (V1.05)
CMD_SET_OTHER_PARAMS changed?	MyMesh.cpp — V3.05 timer logic: next_auto_advert==0 check + immediate advert must be preserved
advert() function changed?	MyMesh.cpp — isEnabled()&&isValid;() AND client_repeat flood logic (V1.03/V2.04)
loop() new blocks?	MyMesh.cpp — timer check must come before dirty_contacts_expiry (V1.00)
writeContactRespFrame() changed?	MyMesh.cpp — type & 0x0F must be preserved (V2.07)
updateContactFromFrame() changed?	MyMesh.cpp — type & 0x0F must be preserved (V2.07)
onChannelMessageRecv() changed?	MyMesh.cpp — V3 !SOS check + return must be preserved; V5 is_favorite call must be preserved (V3.01/V5.03)
sendSOS() still present?	MyMesh.cpp — case-insensitive channel search (V3.01)
sendChannelMessage() still present?	MyMesh.cpp — getCurrentTimeUnique() timestamp (V5.01)
isSenderFavorite() / extractSenderName() still present?	MyMesh.cpp — name search in contact list (V5.03)
repeat_freq_ranges[] still present?	MyMesh.cpp — correct values: 433050/434790, 869400/869650, 915000/928000 (V2.01)
toggleOffGrid(): advert() at end?	MyMesh.cpp — last line before } — RX restart via TX->RX cycle (V3.08)
NodePrefs.h fields still present?	normal_freq/bw/sf/cr at end of struct (V2.02)
DataStore.cpp bytes 90–99?	normal_freq/bw/sf/cr after rx_boosted_gain (byte 89) (V2.02)
DataStore.cpp saveContacts() masking?	V2.06: clean_type = c.type & 0x0F — must be preserved
AbstractUITask.h triggerSOS()?	V3: virtual with empty default — needed for ui-orig compatibility
AbstractUITask.h newMsg() signature?	V5: bool is_favorite = false as last parameter — compiler error without this
setGPSAdvertEnabled() in MyMesh.h?	V3.05: next_auto_advert==0 check + immediate advert() call
HomePage enum correct?	UITask.cpp (ui-new) — SOS after GPS, ADVERT removed, TRACKING between RECENT and RADIO
OutdoorMenuScreen still correct?	UITask.cpp — isOnTrackingPage(), outdoor_menu in begin(), handleLongPress()
SOSAlertScreen + SOSSendScreen?	UITask.cpp — both classes incl. V3.05 _node_name, begin() init, handleLongPress()
AUTO_OFF_MILLIS=0 set?	platformio.ini — in both companion environments (V1.00)
Backlight init fix present?	UITask.cpp — _backlight_initialized guard in loop() DISP_BACKLIGHT block (V4.03)
Buzzer tones in notify() correct?	UITask.cpp — MsgRcv3, kerplop, ack — unchanged from original (V3.01)
variant.h: PIN_BUTTON2 = 39?	V3.07a: if upstream variant.h has corrected this, value is already right

What to check	Why / Where
UITask.cpp: no PIN_BUTTON2 hack at start?	V3.07a: fix belongs in variant.h, not UITask.cpp
UITask.cpp loop(): DISP_BACKLIGHT && !BACKLIGHT_BTN?	V3.07a: avoids conflict with BACKLIGHT_BTN devices
UITask.h: _btn2_was_pressed in #ifdef PIN_BUTTON2?	V3.07a: devices without second button would not compile
UITask.cpp: isOnRecentPage() present?	V3.07c: needed in handleLongPress()
UITask.cpp: isOnFirstPage() present?	V5: needed in handleLongPress() for message screen entry
UITask.cpp: handleLongPress() RECENT branch with _next_refresh=0?	V3.07c: without _next_refresh=0, alert only appears after 5 seconds
UITask.cpp: all showAlert() texts English?	V3.07b: 'SOS sent!', 'No SOS channel!', 'Advert sent!'
V4 layouts correct?	All hint texts at display.height()-13; display.height()>=128 guard; advert_icon_large 48×48
icons.h: advert_icon_large present?	V4.02: 48×48px array, bit-reversed for GxEPD — original advert_icon 32×32 unchanged
V5 screens present?	UITask.cpp: MsgHistoryScreen, MsgFilterScreen, FavPreviewScreen (all); ChannelSelectScreen, MsgComposeScreen (#if UI_HAS_JOYSTICK)
UITask.h: new V5 members present?	_num_unread, _latest_fav_name, _latest_fav_time, setCurrScreen() public, gotoMsgFilter(), gotoChannelSelect(), decrementUnread(), resetUnread()
UITask.cpp: newMsg() updated for V5?	No setCurrScreen(msg_preview); _num_unread++; favorites popup logic; refresh pattern preserved
UITask.cpp: formatRelTime() defined?	Static file-scope helper, used in HomeScreen / MsgHistoryScreen / FavPreviewScreen (V5.09)
ui-orig/Button.h: QUINTUPLE_PRESS in enum?	V5.10: after QUADRUPLE_PRESS, before LONG_PRESS
ui-orig/Button.cpp: _clickCount == 4 (not >= 4)?	V5.10: upstream uses >= 4 which swallows 5x click as quadruple
ui-orig/Button.cpp: _clickCount >= 5 branch?	V5.10: separate branch — triggers QUINTUPLE_PRESS
ui-orig/Button.cpp: QUINTUPLE_PRESS case in triggerEvent()?	V5.10: calls _onQuintuplePress callback
ui-orig/UITask.h: handleButtonQuintuplePress() declared?	V5.10: private method — Off-Grid toggle handler
ui-orig/UITask.h: _sos_active and _buzzer_restore_quiet?	V5.10: both false at init
ui-orig/UITask.h: triggerSOS() override declared?	V5.10: buzzer alarm implementation — not just inherited no-op
ui-orig/UITask.h: refreshDisplay() and isOnRecentOrTrackingPage()?	V5.10: explicit no-op overrides — MyMesh calls both
ui-orig/UITask.h: newMsg() has is_favorite param?	V5.10: bool is_favorite = false — must match AbstractUITask V5
ui-orig/UITask.cpp: onQuintuplePress() callback registered?	V5.10: both _userButton and _userButtonAnalog in begin()
ui-orig/UITask.cpp: _buzzer_restore_quiet restored in loop()?	V5.10: in buzzer else-block after isPlaying()==false
ui-orig/UITask.cpp: SOS siren loop in loop()?	V5.10: _sos_active && !isPlaying() → restart siren
ui-orig/UITask.cpp: short press acknowledges SOS first?	V5.10: handleButtonShortPress() checks _sos_active before display logic

What to check	Why / Where
ui-orig/UITask.cpp: all button feedback forced?	V5.10: 2x/3x/4x/5x all force buzzer.quiet(false) + set _buzzer_restore_quiet
PR #2105 merged?	LED fix for ThinkNode M1 (variant.h/variant.cpp) — then in upstream
PR #2107 merged?	LEDManager refactor across 60+ files — check carefully

10. Operating Guide — ThinkNode M1

Page order (BLE build): FIRST > RECENT > TRACKING > RADIO > GPS > SOS

Page order (USB build): FIRST > RECENT > TRACKING > RADIO > SOS

Lower Button — Normal Operation

Action	Result
1× click	Next page
2× click	Previous page
3× click	Buzzer mute on/off (toggleBuzzer)
Long press — FIRST page	Open MsgFilterScreen
Long press — RECENT page	Send advert immediately + alert 'Advert sent!' (V3.07c)
Long press — TRACKING page	Open Outdoor Settings menu
Long press — SOS page	Open SOS send screen
Long press — GPS page	GPS on/off
Long press — first 8 s after boot	CLI Rescue Mode

Upper Button — Backlight (V4.00)

Action	Result
Short press (backlight off)	Backlight ON + 5 min auto-off timer starts
Short press (backlight on)	Backlight OFF + timer stopped
Lower button pressed (backlight on)	Timer reset to +5 min
5 min without button press	Backlight automatically OFF

Message Filter Screen

Action	Result
Short press	Next menu entry (cyclic: All → Favorites → Back)
Long press	Execute current selection

Message History Screen (M1)

Action	Result
Short press	Next (older) message, decrement unread counter
Double press	Back to MsgFilterScreen
Long press	Clear all messages, return to HomeScreen

Outdoor Settings Menu

Action	Result
Short press	Next menu entry (cyclic)
Long press	Execute current selection (toggle ON/OFF or Back)

SOS Send Screen

State	Action	Result
Idle	Long press	Show confirmation
Confirmation	Long press	Send SOS + return to Home
Confirmation	Short press	Cancel — return to Idle

SOS Alarm Screen (received)

Action	Result
Long press	Acknowledge alarm, stop buzzer, return to Home

Tracking Page — Display Meanings

Display	Meaning
---	Own GPS has no current fix (isEnabled()+isValid() == false)
?	Node sent a GPS advert, but coordinates are 0/0
340m / 1.2km	Line-of-sight distance to node (Haversine calculation)
Settings: long press	Hint pointing to Outdoor Settings menu
Node missing from list	Last advert from this node had no GPS (has_gps=false)

■ *Only contacts marked as favorites are shown (mark in the companion app). Maximum 3 entries. Green = GPS advert less than 5 minutes ago. Yellow = older.*

platformio.ini — Differences BLE vs. USB Environment

Flag	BLE (companion_radio_ble)	USB (companion_radio_usb)
AUTO_OFF_MILLIS=0	Present	Present
PIN_BUZZER=6	Present	Present
NonBlockingRTTTL	Present	Present
ENV_INCLUDE_GPS=1	Present	MISSING (intentional)
BLE_PIN_CODE=123456	Present	MISSING

■ *ENV_INCLUDE_GPS=1 is intentionally absent in the USB environment. Page order USB build: FIRST > RECENT > TRACKING > RADIO > SOS*

11. Operating Guide — Seeed Wio Tracker L1

All L1-specific code is compiled only when UI_HAS_JOYSTICK is defined (platformio_L1.ini). M1 builds are unaffected.

Page order: FIRST > RECENT > TRACKING > RADIO > SOS (no GPS page)

Joystick Navigation

Input	Result
Left/Right (home pages)	Previous / Next page
Short press on FIRST page	Open MsgFilterScreen
Short press on SOS page	GPS on/off (not available on L1 — no GPS)

Message Filter Screen (L1)

Input	Result
Joystick up/down	Navigate menu entries (All / Favorites / Send Message / Back)
Joystick press (short)	Execute current selection
Back button	Return to HomeScreen
Long joystick press	Return to HomeScreen

Message History Screen (L1)

Input	Result
Joystick up/down	Navigate through messages
Long joystick press	Clear all messages, return to HomeScreen
Back button	Return to MsgFilterScreen

Channel Select Screen (L1)

Input	Result
Joystick up/down	Navigate channel list
Joystick press (short)	Select channel, open MsgComposeScreen
Back button / long press	Return to MsgFilterScreen

Message Compose Screen (L1)

Input	Result
Joystick up/down	Move between keyboard rows
Joystick left/right	Move between characters in current row
Short joystick press	Insert selected character (or SPACE / BACKSPACE / SEND)
Long joystick press	Send message
Back button (KEY_BACK)	Cancel, return to ChannelSelectScreen

■ *Keyboard special characters: _ = SPACE, < = BACKSPACE, ~ = SEND*

FavPreview Screen (L1)

Input	Result
Short joystick press	Dismiss, return to HomeScreen
Back button	Dismiss, return to HomeScreen
Long joystick press	Nothing

12. Operating Guide — Headless Devices (T1000-E, RAK WiseMesh TAG)

Headless devices have no display and a single button. All feedback is via buzzer. Ascending tone (c→e) = feature on/enabled. Descending tone (e→c) = feature off/disabled. ui-orig is used for headless devices. ui-new (ThinkNode M1, Wio Tracker L1) is unaffected.

Button Operation

Action	Result	Buzzer
1× press	Wake display (if present) / acknowledge SOS alarm	—
2× press	Send advert (ZeroHop normal, Flood in Off-Grid)	Double beep (c,p,c)
3× press	Buzzer mute on/off	c,e = on / e,c = off
4× press	GPS on/off	c,e = on / e,c = off
5× press	Off-Grid mode on/off	c,e = on / e,c = off
Long press (first 8 s)	CLI Rescue Mode	—
Long press	Shutdown	—

SOS Alarm (received)

Situation	Behaviour
!SOS message received	Siren starts immediately, loops continuously
1× press	Alarm acknowledged — siren stops
Long press during alarm	Shutdown (long press always = shutdown on headless)

Notes

■ *Off-Grid state is saved to flash and restored on reboot.*

■ *All button feedback tones are played regardless of buzzer mute state.*

■ *SOS send is not available on headless devices — no free button slot for two-step confirmation.*

■ *The advert sent by 2× press follows the normal advert() logic: ZeroHop in normal mode, Flood in Off-Grid mode.*

■ *GPS toggle (4×) only has effect on devices with GPS hardware (T1000-E has GPS; RAK WiseMesh TAG may vary).*

10. V5.11 — BLE App-Init Race Condition Fix

Changed files: MyMesh.h, MyMesh.cpp

After BLE connect, the onSecured() callback fired and immediately pushed contacts, advert paths and the offline queue to the app before the CMD_APP_START handshake was complete. On slow BLE connections this caused the app to time out during initial sync.

Root cause: all async push functions checked only _serial->isConnected() — not whether the handshake had been acknowledged.

```
// MyMesh.h:
bool _app_initialized = false;

// MyMesh.cpp — onSecured():
_app_initialized = false;  // reset — handshake not yet complete

// MyMesh.cpp — CMD_APP_START handler (end of block):
_app_initialized = true;   // handshake complete, async pushes now allowed

// All async push locations gated:
if (_app_initialized && _serial->isConnected() && ...) { /* push */ }
```

- onLogout() / onDisconnect() reset _app_initialized = false.

11. V5.12 — Favourite Detection Regression Fix

Changed file: MyMesh.cpp

Two bugs caused favourite detection to fail for channel messages after V5.10.

Bug 1 — Headless merge side effect: During V5.10 headless integration the newMsg() call in onChannelMessageRecv() was not fully re-checked after the signature change. The is_favorite flag was computed but not reliably passed in all code paths.

Bug 2 — extractSenderName pipe-delimiter: The original implementation used ' | ' as separator ('NodeName | Region: text'). Nodes whose names contain ' | ' (e.g. 'Juno48 | EU') were split incorrectly — isSenderFavorite() found no match.

Fix: extractSenderName() rewritten to use ': ' as delimiter.

```
static bool extractSenderName(const char* text, char* out_name, int max_len) {
  const char* colon = strstr(text, ": ");
  if (colon == NULL) return false;
  int len = (int)(colon - text);
  if (len <= 0 || len >= max_len) return false;
  strncpy(out_name, text, len);
  out_name[len] = 0;
  return true;
}
```

- MeshCore format is always 'NodeName: text'. Node names may contain ' | ' but never ': ' — so ': ' is an unambiguous separator.

12. V5.13 — App-Sync Clears Favourite Cache

Changed file: ui-new/UITask.cpp

When the app connected and synced all messages (msgRead(0)), the MSG counter reset correctly but the favourite contact line on the FIRST page still showed the stale name and timestamp from the previous session. After disconnect the stale info reappeared, creating a misleading UI state.

```
void UITask::msgRead(int msgcount) {
  _msgcount = msgcount;
  if (msgcount == 0) {
    ((MsgHistoryScreen*) msg_history)->clearAll();  // resets _num_unread
    _latest_fav_name[0] = 0;  // V5.13: clear favourite cache on app sync
    _latest_fav_time    = 0;
    gotoHomeScreen();
  }
```

```
}
```

13. V5.2 — HomeScreen Redesign & Online Node Counter

Changed files: ui-new/UITask.cpp, ui-new/UITask.h, ui-new/icons.h, MyMesh.cpp, AbstractUITask.h, ui-orig/UITask.h

Overview of changes

Area	Change
HomeScreen FIRST	Complete rebuild — see layout tables below
RECENT page	Adaptive row limit: max_y = height>=128 ? 53 : 42 — prevents footer overlap on 64px OLED
TRACKING page	Same adaptive row limit as RECENT
newMsg() signature	channel_name added as last param (const char*, default nullptr) — stored in MsgHistoryScreen for accurate channel tag
Online node cache	_online_nodes[16] ring buffer in UITask; updateOnlineNode() + countOnlineNodes() + countOnlineNodesExcluding() track message senders (in addition to advert senders)
icons.h	nodes_icon 8×8px filled-circle XBM added — symmetric bytes, no bit-reversal needed for either display driver
AbstractUITask.h	newMsg() signature: channel_name = nullptr added as last parameter
ui-orig/UITask.h	newMsg() signature updated to match — channel_name intentionally unused in headless path
MyMesh.cpp	updateOnlineNode() called in queueMessage() and onChannelMessageRecv(); channel_name passed to newMsg()

HomeScreen FIRST — Large display (M1, height ≥ 128px)

Y pos	Element	Condition
y=0	Node name (left) + battery indicator (right)	always
y=14	Page indicator dots	always
y=22	'< Connected >' (size 1) OR 'MSG: X' (size 2)	connected / disconnected
y=33	nodes_icon 8×8 + ' X nodes' (online counter)	always
y=44	Mode line: '[GPS-SHARE] [OFF-GRID]'	if any mode active; advances y+=11
y=44+	Favourite line: '* Name Xm'	disconnected AND favourite message exists
h-13	Footer hint: 'Messages: longpress'	always

HomeScreen FIRST — Small display (L1, height < 128px)

Y pos	Element	Condition
y=22	'MSG: X' (left, size 1)	disconnected only
y=22	nodes_icon 8×8 + ' X' (right-aligned)	always
y=33	Priority: (1) '< Connected >' (2) '* Name Xm' (3) mode '[GPS-SHARE][OG]' (4) empty	first matching condition
h-13	Footer hint: 'Messages: press Enter'	always

Online node counter — architecture

The counter combines two sources to avoid double-counting nodes heard in the last 30 minutes:

(1) Advert senders — from advert_paths[] via getRecentlyHeard() using the full ADVERT_PATH_TABLE_SIZE buffer. Not UI_RECENT_LIST_SIZE — that 4-slot limit was the V5.10 bug that capped the counter at 4.

(2) Message-only senders — nodes that sent a DM or channel message without advertising. Tracked in _online_nodes[16] in UITask. countOnlineNodesExcluding() skips entries already covered by a fresh

advert.

```
// UITask.h — new members:
struct OnlineNode { char name[33]; uint32_t last_seen; };
static const int ONLINE_CACHE_SIZE = 16;
OnlineNode _online_nodes[ONLINE_CACHE_SIZE];

// Public methods:
void updateOnlineNode(const char* name, uint32_t timestamp);
int  countOnlineNodes(uint32_t now, uint32_t window_secs = 1800) const;
int  countOnlineNodesExcluding(uint32_t now, uint32_t window_secs,
                               const AdvertPath* adv_buf, int adv_count) const;

// MyMesh.cpp — updateOnlineNode() call sites:
// queueMessage() (direct messages):
_ui->updateOnlineNode(from.name, getRTCClock()->getCurrentTime());

// onChannelMessageRecv():
_ui->updateOnlineNode(display_name, getRTCClock()->getCurrentTime());
```

■ updateOnlineNode() is called regardless of connection state — the counter reflects all heard nodes, not just off-grid messages.

newMsg() signature update

```
// AbstractUITask.h:
virtual void newMsg(uint8_t path_len, const char* from_name, const char* text,
                    int msgcount, bool is_favorite = false,
                    const char* channel_name = nullptr) = 0;

// ui-orig/UITask.h — must match (channel_name intentionally unused):
void newMsg(uint8_t path_len, const char* from_name, const char* text,
            int msgcount, bool is_favorite = false,
            const char* channel_name = nullptr) override;
```

14. Checklist Additions — V5.11 to V5.2

Add these entries to the checklist (Section 9 of base documentation):

What to check	Why / Where
MyMesh.h: _app_initialized member?	V5.11: bool, init false — BLE race condition fix
MyMesh.cpp: _app_initialized=false in onSecured()?	V5.11: reset before handshake
MyMesh.cpp: _app_initialized=true in CMD_APP_START?	V5.11: set only after handshake complete
MyMesh.cpp: all async pushes gated on _app_initialized?	V5.11: prevents queue flood on slow BLE
extractSenderName() uses ': ' not ' | '?	V5.12: node names may contain ' | '; ': ' is unambiguous
msgRead(): fav cache cleared on msgcount==0?	V5.13: _latest_fav_name[0]=0, _latest_fav_time=0
AbstractUITask.h: newMsg() has channel_name param?	V5.2: const char* channel_name = nullptr — compiler error without this
ui-orig/UITask.h: newMsg() signature matches?	V5.2: channel_name = nullptr — intentionally unused in headless path
UITask.h: OnlineNode struct + _online_nodes[16]?	V5.2: message-sender tracking cache
UITask.h: updateOnlineNode(), countOnlineNodes(), countOnlineNodesExcluding()?	V5.2: public methods for online node tracking
UITask.cpp: updateOnlineNode() called in newMsg()?	V5.2: from_name passed after storing message
MyMesh.cpp: updateOnlineNode() in queueMessage() AND onChannelMessageRecv()?	V5.2: called regardless of connection state
HomeScreen FIRST: uses ADVERT_PATH_TABLE_SIZE buffer (not UI_RECENT_LIST_SIZE)?	V5.2: old code capped counter at 4
HomeScreen FIRST: countOnlineNodesExcluding() (not countOnlineNodes)?	V5.2: prevents double-counting advert+message senders
HomeScreen FIRST: MSG:X hidden when connected?	V5.2: if (!connected) guard around MSG:X block
icons.h: nodes_icon 8×8 filled circle present?	V5.2: symmetric XBM bytes — no bit-reversal needed
RECENT page: max_y = height>=128 ? 53 : 42?	V5.2: adaptive row limit, prevents footer overlap
TRACKING page: same max_y adaptive limit?	V5.2: identical pattern to RECENT page